LABORATORY · Lab · 26 hours

Experiment 6: Frequency Domain Analysis Using Bode Plot



Control Systems Laboratory (22EEG53)

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Learning objectives

- Plot the Bode magnitude and phase of a transfer function
- Determine gain margin, phase margin and crossover frequencies
- Assess stability from the margins

Syllabus context

Laboratory component of 22EEG53 Control Systems (0:0:2 hours per week). CIE: record and observation 25, viva-voce 5, lab test 20 marks.

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Aim and requirements

Aim: to perform frequency domain analysis using the Bode plot in MATLAB and determine GM and PM.

Software: MATLAB with the Control System Toolbox (Simulink where required).

Hardware: Personal computer.

System: G(s) = 2/[s(s + 1)(s + 2)] (unity feedback).

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Theory

Bode plot

Magnitude $20 \log|G(j\omega)|$ and phase vs $\log \omega$.

Gain margin

$GM = -20 \log|G(j\omega_{pc})|$ at $\angle G = -180°$.

Phase margin

$PM = 180° + \angle G(j\omega_{gc})$ at $|G| = 1$.

Stable if GM > 0 dB and PM > 0°.

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Procedure

1

Define G

G = tf(2, conv([1 0], conv([1 1],[1 2]))).

2

Bode plot

bode(G); grid on.

3

Margins

margin(G) marks GM, PM and the crossover frequencies.

4

Interpret

Check the closed loop: step(feedback(G,1)).

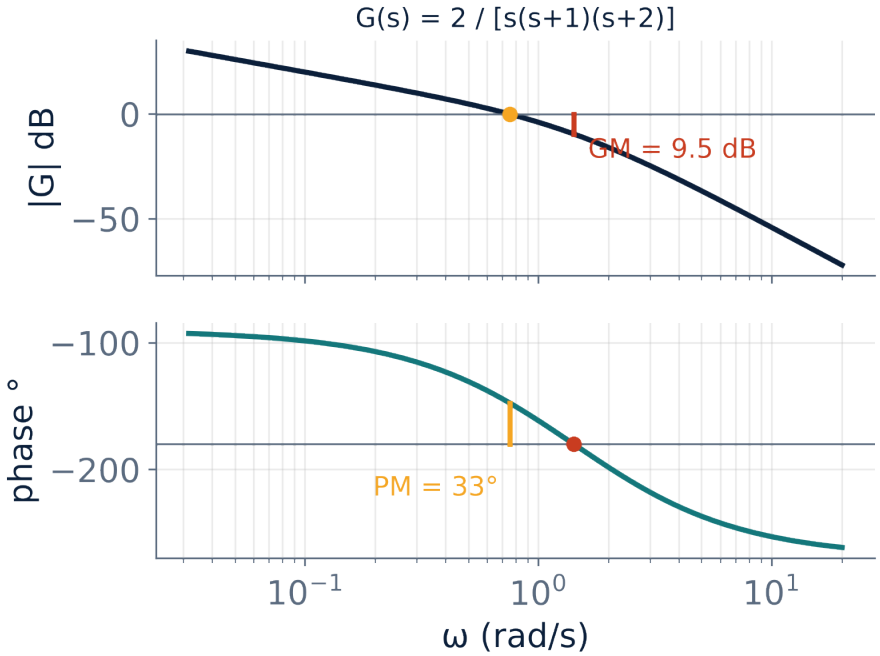
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MATLAB program

```
G = tf(2, conv([1 0], conv([1 1], [1 2])));
figure, bode(G), grid on
figure, margin(G)
[Gm, Pm, wpc, wgc] = margin(G);
fprintf('GM = %.2f dB at %.3f rad/s\n', 20*log10(Gm), wpc)
fprintf('PM = %.1f deg at %.3f rad/s\n', Pm, wgc)
figure, step(feedback(G, 1))      % closed loop response
```

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Expected result



- GM ≈ 9.54 dB at ωpc = 1.414 rad/s.
- PM ≈ 32.7° at ωgc = 0.748 rad/s.
- Both positive: stable.

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Observation table

Gain K	GM (dB)	PM (deg)	ωgc	Stability
1				
2	9.54	32.7	0.748	Stable
6	0	0	1.414	Marginal
10				

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Key takeaways

- bode() and margin() give the frequency response and margins
- GM = 9.54 dB, PM = 32.7° for K = 2
- K = 6 gives GM = 0 dB: marginal stability
- Larger K reduces the margins

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Practice questions

Q1. Define GM and PM.

Q2. What are the margins for K = 6?

Q3. Write the MATLAB commands used.

Q4. Predict the closed-loop overshoot from PM = 32.7°.